

JEE-Main-23-01-2025 (Memory Based) [EVENING SHIFT]

Physics

Question: Two charged particles have values $14 \mu\text{C}$ & $-14 \mu\text{C}$ are at positions $(7, 0, 0)$ m & $(-7, 0, 0)$ m. Find potential energy between them .

Options:

- (a) 0.126 J
- (b) -0.126 J
- (c) 0.252 J
- (d) -0.252 J

Answer: (b)

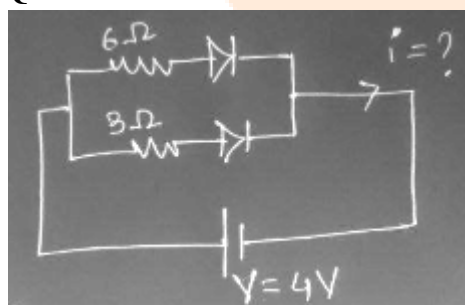
Question: If 40 m takes 27 days to revolve around earth, then the time taken by an object to revolve around earth which is at a distance of $1/9$ times the distance of the moon from Earth is

Options:

- (a) 1 day
- (b) 9 days
- (c) 3 days
- (d) 27 days

Answer: (a)

Question: Diodes are ideal. Find current through the battery

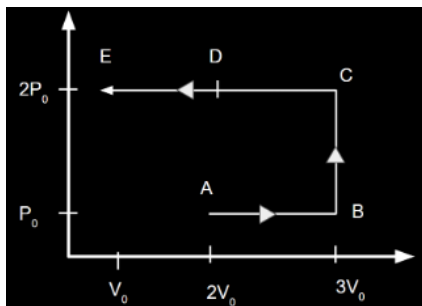


Options:

- (a) 1 A
- (b) 2 A
- (c) 3 A
- (d) 4 A

Answer: (b)

Question: Find total work done from A to E



Options:

- (a) $-3P_0V_0$
- (b) $3P_0V_0$
- (c) $2P_0V_0$
- (d) $5P_0V_0$

Answer: (a)

Question: A disc of mass M and Radius R is rotating about its axis. If the angle rotated about it as a function of time ' t ' is $\theta = at^2 + bt + c$ where a , b and c are constants then find the power derived to the disc as a function of time .

Options:

- (a) $aMR^2 (2at + b)$
- (b) aMR^2
- (c) $2 a^2MR^2 t$
- (d) $aMR^2 b$

Answer: (a)

Question: A convex mirror of focal length ' f ' is dipped in a liquid of refraction index μ . The new focal length is

Options:

- (a) $2f$
- (b) μf
- (c) f
- (d) f/μ

Answer: (c)

Question: A projectile is thrown at an angle of 60° . It has kinetic energy K . What will be its kinetic energy at the top most position ?

Options:

- (a) $2K$
- (b) $4K$
- (c) $K/2$
- (d) $K/4$

Answer: (d)

Question: The work function of a metal is 2.14 eV. Find the wavelength of the incident light if the stopping potential of 0.6 V.

Options:

- (a) 326 nm
- (b) 454 nm
- (c) 524 nm

(d) 232 nm

Answer: (b)

Question: Statement 1: Graph of frequency f of X ray and atomic number Z of heavy nucleus is straight line, in

X ray emission.

Statement 2: Graph of square root of frequency $\sqrt{f}$ of X ray and atomic number Z of heavy nucleus

is straight line, in X ray emission.

Options:

(a) Statement 1 is correct and statement 2 is correct

(b) Statement 1 is incorrect and statement 2 is correct

(c) Statement 1 is correct and statement 2 is incorrect

(d) Statement 1 is incorrect and statement 2 is incorrect

Answer: (b)

Question: In an LCR circuit, inductor is 100 mH, capacitor is 25 nF, resistor is 10 k Ω . Find angular freqn at which current becomes maximum

Options:

(a) 2×10^4 rad/s

(b) 2×10^3 rad/s

(c) 2×10^2 rad/s

(d) 2×10^5 rad/s

Answer: (a)

Question: A prism of refractive index of $\mu = \sqrt{2}$ is placed such that angle prism is equal to minimum angle of deviation. Find angle of prism.

Options:

(a) 60°

(b) 30°

(c) 45°

(d) 15°

Answer: (a)

Question: Find the pressure differences of air bubble of radius 2 cm formed 20 cm below an open water surface and atmospheric pressure

(given surface tension of water = 70×10^{-3} N/m)

Options:

(a) 0.3×10^5 N/m²

(b) 0.022×10^5 N/m²

(c) 2.03×10^3 N/m²

(d) 1.03×10^5 N/m²

Answer: (b)

Question: A capacitor of 0.25 μ F connected across a variable voltage has a displacement current of 2.5 mA at a certain moment. Find the rate of change of voltage across capacitor at this moment

Options:

(a) 10000 Volt/sec

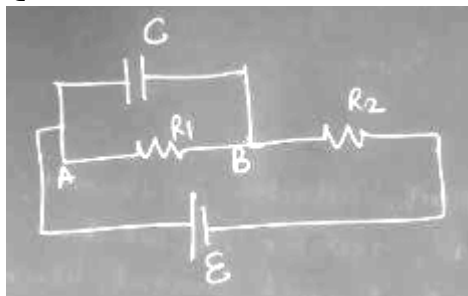
(b) 1000 Volt/sec

(c) 100 Volt/sec

(d) 10 Volt/sec

Answer: (a)

Question:



Find q on capacitor in steady state .

Options:

(a) $c\epsilon$

(b) $c\epsilon/2$

(c) $2c\epsilon$

(d) $c\epsilon/4$

Answer: (b)

Question: Statement 1 :For $30 < A < 170$, the binding energy per nucleon is almost constant

Statement 2 : Nuclear force is long range force

Options:

(a) Both the Statements are correct

(b) Only Statement 2 is correct

(c) Only Statement 1 is correct

(d) Both the statements are wrong

Answer: (c)

Question: There is a horizontal pipe of variable cross-section having fluid of density flowing through it. At cross section A & B the velocities are V_A & V_B and pressure P_A & P_B . Find the correct relation between velocities.

Options:

(a) $V_A - V_B = \frac{\rho}{2(P_B^2 - P_A^2)}$

(b) $V_A - V_B = \frac{2(P_A - P_B)}{\rho}$

(c) $V_A^2 - V_B^2 = \frac{2(P_B - P_A)}{\rho}$

(d) $V_A^2 - V_B^2 = \frac{2(P_A - P_B)}{\rho}$

Answer: (c)

Question: In an EM Wave of Frequency 20 Mhz Electric Field is 9.3 V/m. What is the Value of Magnetic Field at that instant?

Options:

- (a) $3.1 \times 10^8 \text{ T}$
- (b) $3.1 \times 10^{-8} \text{ T}$
- (c) $0.13 \times 10^8 \text{ T}$
- (d) $0.13 \times 10^{-8} \text{ T}$

Answer: (b)

Question: If in a rod tension of 7 N produces extension x_1 & tension of 5 N produces extension x_2 then find tension that produces extension $7x_1 - 5x_2$

Options:

- (a) 50 N
- (b) 24 N
- (c) 36 N
- (d) 128 N

Answer: (b)



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[EVENING SHIFT]

Chemistry

Question: The correct order of melting point of 14 group element is

Options:

- (a) $C > Si > Ge > Sn > Pb$
- (b) $Si > C > Ge > Sn > Pb$
- (c) $Ge > Sn > C > Si > Pb$
- (d) $C > Si > Ge > Pb > Sn$

Answer: (d)

Question: What will be effect on pH of water when it is heated

Options:

- (a) Increase
- (b) Decrease
- (c) Remains same
- (d) pH first increases then decreases

Answer: (b)

Question: α -helix protein and β -pleated sheet protein belong from which of the following structures?

Options:

- (a) Primary
- (b) Secondary
- (c) Tertiary
- (d) Quaternary

Answer: (b)

Question: Match the following List-I with List-II

	List-I (Alloys)		List-II (Metals)
A.	Bronze	(i)	Fe, Cr, and Ni
B.	Stainless steel	(ii)	Cu and Sn
C.	UK Gold Coin	(iii)	Cu and Zn
D	Brass	iv	Ag, Cu, Zn and Ni

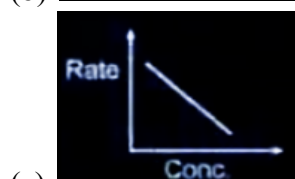
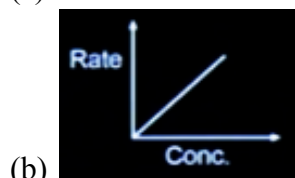
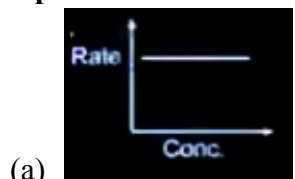
Options:

- (a) A-ii, B-i, C-iv, D-iii
- (b) A-iii, B-iv, C-i, D-ii
- (c) A-iv, B-iii, C-ii, D-i
- (d) A-i, B-ii, C-iii, D-iv

Answer: (a)

Question: Which one the following plots represents zero order reactions

Options:



Answer: (a)

Question: By using relation

$$\Delta G = \Delta H - T\Delta S$$

Which of the following is incorrect for spontaneous reaction at a given temperature

Options:

(a) $\Delta H > 0$, $\Delta S > 0$

(b) $\Delta H > 0$, $\Delta S < 0$

(c) $\Delta H < 0$, $\Delta S > 0$

(d) $\Delta H < 0$, $\Delta S < 0$

Answer: (b)

Question: Statement-I: For a particular shell, maximum number of orbital is n^2 .

Statement-II: For d-subshell, the number of orientations lies between -1 to +1 including zero.

Options:

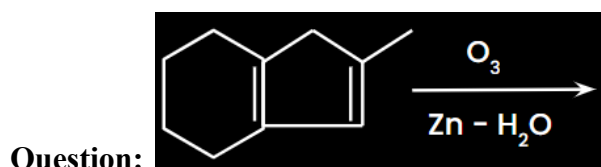
(a) S-I and S-II both are correct

(b) S-I and S-II both are incorrect

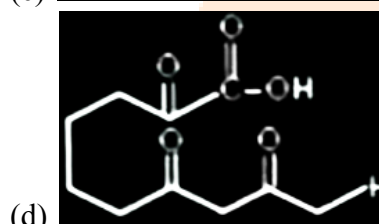
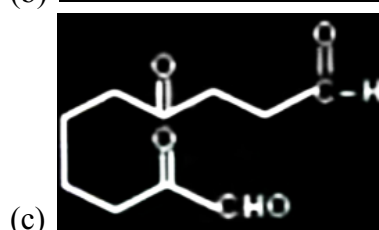
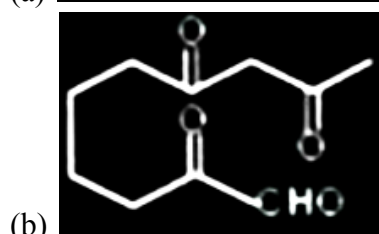
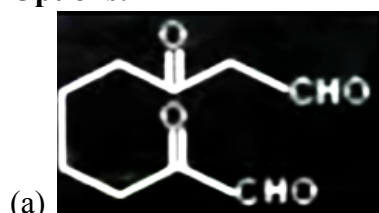
(c) S-I is correct, S-II is incorrect

(d) S-I is incorrect, S-II is correct

Answer: (a)



Options:



Answer: (b)

Question: The total Number of isomers possible (aldehyde and Ketone) for C_4H_8O are

Options:

- (a) 6
- (b) 3
- (c) 4
- (d) 5

Answer: (b)

Question: Identify the complex in which central metal ion has d^4 configuration

Options:

- (a) $[FeO_4]^{2-}$
- (b) $[Mn(CN)_6]^{3-}$
- (c) $[Fe(CN)_6]^{3-}$
- (d) $[NiF_6]^{2-}$

Answer: (b)

Question: Consider the given following reaction $X_2Y(S) \rightleftharpoons X_2(g) + \frac{1}{2}Y_2(g)$. If α is the degree of dissociation. Calculate K_p in terms of P total pressure

Options:

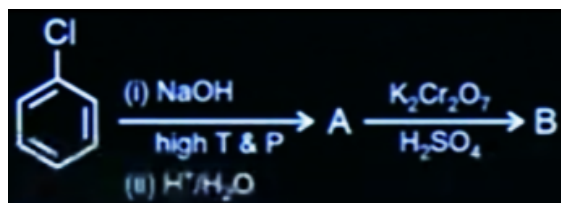
(a) $K_p = \frac{2P^{\frac{3}{2}}}{3^{\frac{3}{2}}}$

$$(b) K_p = \frac{2P^{\frac{3}{2}}}{3}$$

$$(c) K_p = \sqrt{\frac{2P}{3}}$$

$$(d) K_p = \frac{\sqrt{2}P}{3}$$

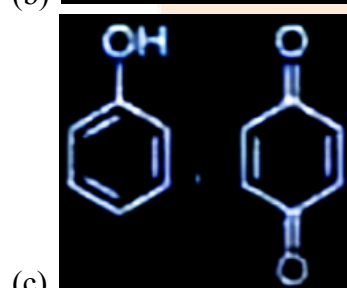
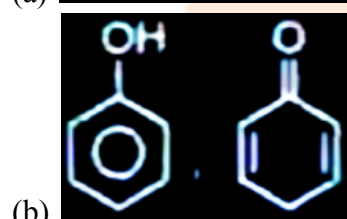
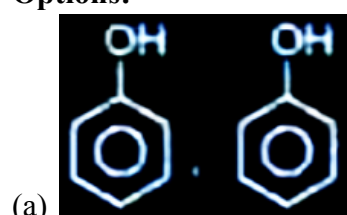
Answer: (a)



Question:

Predict A and B?

Options:



(d) None of the above

Answer: (c)

Question: Consider the following E° values of given half cell

$$E^\circ_{Ag^+/Ag} = 0.8 V, E^\circ_{Zn^{2+}/Zn} = -0.76 V$$

$$E^\circ_{Cu^{2+}/Cu} = 0.34 V, E^\circ_{Mg^{2+}/Mg} = -2.36 V$$

Then which of the following will have the most negative value of ΔG° ?

Options:

- (a) $Zn|Zn^{2+}||Cu^{2+}|Cu$
- (b) $Mg|Mg^{2+}||Ag^{2+}|Ag$
- (c) $Mg|Mg^{2+}||Zn^{2+}|Zn$
- (d) $Cu|Cu^{2+}||Ag^{2+}|Ag$

Answer: (b)

Question: When a non solute (A) is added to a volatile solvent, the vapour pressure of solvent decrease by 10 mm Hg. Mole fraction of solute is 0.2. If 2nd solute (B) is added to the same solution and vapour pressure of solution decreases by 20 mm Hg. Calculate mole fraction of 2nd solute in the final solution.

Options:

- (a) 0.3
- (b) 0.4
- (c) 0.5
- (d) 0.6

Answer: (c)

Question: Given below are two statements:

Statement-I: Phenol & alcohol melting point increases with increase in carbon atoms

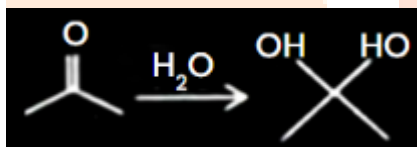
Statement-II: Phenol & alcohol has higher melting point than ether and haloalkanes

In the light of the above statements, choose the most appropriate answer from the options given below:

Options:

- (a) Both Statement-I and Statement-II are incorrect below
- (b) Statement-I is correct but statement-II is incorrect
- (c) Both Statements are correct
- (d) Statement I is incorrect and Statement II is correct

Answer: (c)



Question:

Given below are two statements:

Statement-I: HCHO ($r = 22400$) rate due to less steric.

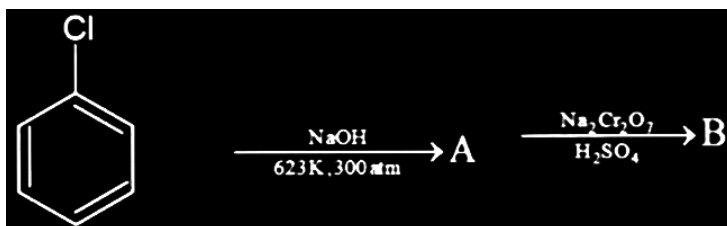
Statement-II: $\text{CCl}_3 - \text{CHO}$ ($r = 2000$) rate due to -I group of halogen

In the light of the above statements, choose the most appropriate answer from the options given below:

Options:

- (a) Both Statement-I and Statement-II are incorrect
- (b) Statement-I is correct & statement-II is incorrect
- (c) Both Statements are correct
- (d) Statement I is incorrect and Statement II is correct

Answer: (d)



Question:

Find B?

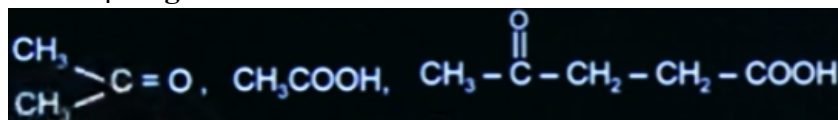
Options:

- (a) 1,2-dimethyl benzene

- (b) Toluene
- (c) Phenol
- (d) 1,4-Benzoquinone

Answer: (d)

Question: A compound X consumes two moles of H_2 and when 'X' heated with $KMnO_4/H^+$ give



The number of σ bonds in X are ____.

Answer: (27)

Question: 81 g of Al reacts with 128 g of O_2 . Calculate the amount(in moles) of Al_2O_3 is produced?

Answer: (1.5)

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[EVENING SHIFT]

Math

Question: In AP with $cd = \frac{3}{2} : 3x^2 - px + q$ was the equation and their roots are 10th and

11th terms and sum of 11 terms is 88 find $q - 2p$.

Solution:

$$\frac{11}{2} \left[2a + 10 \times \frac{3}{2} \right] = 88$$

$$2a + 15 = 16$$

$$2a = 1$$

$$T_{10} = \frac{1}{2} + \frac{27}{2} = 14 \quad T_{11} = \frac{31}{2}$$

$$\frac{p}{3} = \frac{59}{2} \quad \frac{q}{3} = \frac{14 \times 31}{2}$$

$$q - 2p = \frac{42 \times 31}{2} - 3 \times 59$$

$$= 474$$

Question: f a square is divided in 4×4 squares. If two squares are chosen randomly then the probability that the squares doesn't share common side is

Options:

- (a) $\frac{3}{5}$
- (b) $\frac{4}{5}$
- (c) $\frac{3}{20}$
- (d) $\frac{10}{7}$

Answer: (b)

Total squares = 16

$$\text{Chosen} = {}^{16}C_2 = \frac{16 \times 15}{2} = 120$$

Adj square : Horizontal row

$$4 \times 3 = 12$$

Vertical : 4×72

$\therefore$ Total adjacent = 24

$$\therefore P(E) = \frac{24}{120} = \frac{1}{5}$$

$$\therefore P(E) = \frac{4}{5}$$

No. of pair squares which share a common side = 24 = (12 × 12)

No. of squares which don't share = ${}^{16}C_2 - 24$

$$= 120 - 24 = 96$$

$$\therefore \text{Prob} = \frac{96}{120} = \frac{8}{10} = \frac{4}{5}$$

Question: There are 5 boys and 4 girls. The sum of number of ways to sit them such that all boys sit together and number of ways such that no boys sit together is equal to

Solution :

$$(2) 5! \times 5! + 4! \times {}^5C_5 \times 5!$$

$$= 120^2 + 24 \times 120 = 144 \times 120$$

$$= 17280$$

Question: Find the variance of numbers 8, 21, 34, ..., 320.

Solution :

8, 21, 34,320

$$T_n = (13n - 5)$$

Variance of 1, 2,25

$$= \frac{\sum n^2}{25} - \left(\frac{\sum n}{25} \right)^2$$

$$= \frac{25 \times 26 \times 51}{6 \times 25} - \left(\frac{25 \times 26}{2 \times 25} \right)^2$$

$$= 13 \times 17 - 13^2 = 13 \times 14 = 52$$

$$\text{so reg var} = 13^2 \times 52$$

$$= 8788$$

$$\begin{aligned}\text{variance} &= \left(\frac{n^2-1}{12}\right)d^2 \\ &= \frac{25^2-1}{12} \times 13^2 \\ &= 52 \times 169 = 8788\end{aligned}$$

Question: Let S be the region consisting of points (x,y) such that

$$-1 \leq x \leq 1 \text{ \& } 0 \leq y \leq a + e^{|x|} - e^{-|x|} \text{ if area bounded by region } 2\left(\frac{e^2 + 8e + 1}{e}\right) \text{ is find "a".}$$

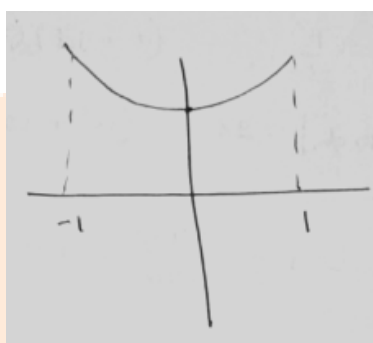
Solution :

$$-1 \leq x \leq 1$$

$$y = a + e^{|x|} - e^{-|x|}$$

$$\begin{aligned}A &= 2 \int_0^1 a + e^x - e^{-x} dx \\ &= 2[a + (e - 1) + (e^{-1} - 1)] \\ &= 2\left[\frac{e^2 + e(a-2) + 1}{e}\right]\end{aligned}$$

$$\text{So } a = 10$$



Question: Shortest distance between P(0, a) and parabola $y^2 = 4x$ is 4. Find the equation of circle whose center lies on axis of parabola and passing through P and focus of parabola.

Solution :

$$y = mx - 2m - m^3$$

$$0 = ma - 2m - m^3$$

$$m=0 \quad m^2 = (a-2)$$

$$(a-m^2)^2 + (2m)^2 = 16$$

$$4 + 4m^2 = 16$$

$$m^2 = 3$$

$$a = 5$$

Eq of Circle

$$(x-1)(x-5) + y^2 = 0$$

$$x^2 + y^2 - 6x + 5 = 0$$

$$y = mx - 2m - m^3$$

$$0 = ma - 2m - m^3$$

$$m = 0 \quad m^2 = (a - 2)$$

$$(a - m^2)^2 + (2m)^2 = 16$$

$$4 + 2m^2 = 16$$

$$m^2 = 6$$

$$a = 8$$

Eq of circle

$$(x - 1)(x - 8) + y^2$$

$$x^2 + y^2 - 9x + 8$$

Question: Find length of chord whose midpoint is $\left(1, \frac{1}{2}\right)$ of ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 1$.

Solution :

$$\frac{x-1}{\cos \theta} = \frac{y-\frac{1}{2}}{\sin \theta} = r$$

$$\frac{(r \cos \theta + 1)^2}{2} + \frac{(r \sin \theta + \frac{1}{2})^2}{4} = 1$$

$$\text{sum} = 0 \rightarrow \left[\cos \theta + \frac{\sin \theta}{4} \right] = 0 \rightarrow \tan \theta = -4$$

$$\text{Pr o.} = \frac{\left| \frac{1}{2} + \frac{1}{16} - 1 \right|}{\frac{\cos^2 \theta}{2} + \frac{\sin^2 \theta}{4}} = \frac{7}{16 \left[\frac{1}{2 \cdot 17} + \frac{16}{17 \cdot 4} \right]}$$

$$= \frac{7 \times 17}{8 + 64} = \frac{7 \times 17}{72} = r_1^2$$

$$\text{Length} = 2r_1 = 2\sqrt{\frac{119}{72}} = \sqrt{\frac{119}{18}}$$



Question: If the square of the shortest distance between the lines $\frac{x-2}{1} = \frac{y-1}{2} = \frac{x+3}{-3}$

$\frac{x+1}{2} = \frac{y+3}{4} = \frac{x+5}{-5}$ is $\frac{m}{n}$, and where m, n are coprime numbers then m + n is

equal to

Options:

(a) 6

(b) 9

(c) 14

(d) 21

Answer: (b)

$$S.D. = \frac{\begin{vmatrix} 3 & 4 & 2 \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}$$

$$= \frac{16-41}{2\hat{i} - \hat{j} + 0\hat{k}} = \frac{2}{\sqrt{5}}$$

$$\frac{m}{n} = \frac{4}{5} \rightarrow m + n = 9$$

Question: If $I = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x dx}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}$, then the value of definite integration $\int_0^{2\pi} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$ is

Options:

(a) $\frac{\pi}{16}$

(b) $\frac{\pi^2}{16}$

$$(c) \frac{\pi}{8}$$

$$(d) \frac{\pi^2}{8}$$

Answer: (b)

$$I = \int_0^{\pi/2} 1 \cdot dx$$

$$2I = \int_0^{\pi/2} 1 \cdot dx \quad I = \frac{\pi}{4}$$

$$I_1 = \int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$I_1 = \int_0^{\pi/2} \frac{\frac{\pi}{2} \sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$I_1 = \frac{\pi}{2} \int_0^{\pi/4} \frac{\tan x \cdot \sec^2 x}{1 + \tan^4 x} dx$$

$$= \frac{\pi}{2} \left[\frac{\tan^{-1}(\tan^2 x)}{2} \right]_0^{\pi/4} = \frac{\pi}{2} \left[\frac{\pi}{8} \right]$$

$$= \frac{\pi^2}{16}$$

Question: In expansion of $(1+x)^p (1-x)^q$ coefficient of x and x^2 is 1 and -2 then find $p^2 + q^2$.

Solution :

$$(1+x)^p (1-x)^q = (1 + px + {}^p C_2 x^2) (1 - qx + {}^q C_2 x^2)$$

$$p - q = 1$$

$$-pq + \frac{p(p-1)}{2} + \frac{q(q-1)}{2} = -2$$

$$(p^2 + q^2 - 2pq) - p - q = -4$$

$$p + q = 5$$

$$p = 3 \quad q = 2$$

$$p^2 + q^2 = 13$$

Question: If $y = \left(x - y \frac{dy}{dx}\right) \sin\left(\frac{x}{y}\right)$ if $x(1) = \frac{\pi}{2}$ then find $\cos(x(2))$.

Solution :

$$x = vy$$

$$y = \left[vy - y\left(v + y \cdot \frac{dv}{dy}\right)\right] \sin v$$

$$1 \cdot \cos v = v - v - y \frac{dv}{dy}$$

$$\frac{dy}{y} + \sin v dv$$

$$\ln y - \cos v = c$$

$$\ln y - \cos \frac{x}{y} = c$$

$$\ln 1 - \cos \frac{\pi}{2} = c = 0$$

$$\ln 2 - \cos \frac{x}{2} = 0$$

$$\cos\left(\frac{x}{2}\right) = \ln 2$$

$$\cos(x) = 2\cos^2 \frac{x}{2} - 1$$

$$= 2(\ln 2)^2 - 1$$

Question: $A = \{(x, y) \mid |x + y| \geq 3\}$;

$B = \{(x, y) \mid |x| + |y| \leq 3\}$

Let $C = A \cap B$. Find the sum of $x + y \forall x, y \in C$.

Solution :

$$A = |x + y| \geq 3 \rightarrow x + y \geq 3 \text{ or } x + y \leq -3$$

$$B = |x| + |y| \leq 3$$

